Yiqun Cai

yiquncai1118@gmail.com | Research Portfolio

Research Interests

Precision analog circuits, low-noise measurement, and sensor interfaces; diffractive waveguides, AR near-eye displays, and optical system fabrication

Education

Institute of Science Tokyo (formerly Tokyo Institute of Technology) Sep. 2026 – Present

Ph.D. Student, Electrical and Electronic Engineering

University of Florida Aug. 2022 – May 2024

M.S., Electrical and Computer Engineering | GPA: 3.66/4.00

Outstanding International Student Scholarship (awarded three times)

Xidian University Sep. 2018 – Jun. 2022

B.Eng., Electronic Science and Technology | GPA: 3.19/4.00

Professional Experience

UEAscend | Hardware Engineer | Shanghai, China Jul. 2024 – Aug. 2026

- Contributed to the design and comparison of two PT1000/thermistor acquisition boards: an AD8221-based Wheatstone bridge and a dual-AD7177 ADC architecture using bidirectional current excitation and a precision reference resistor. The dual-ADC scheme combines four readings per sensor to suppress thermoelectric-EMF and ADC-to-ADC errors.
- Rebalanced bridge resistances and amplifier gain to extend sensor compatibility from PT1000 (~1 kΩ) to higher-resistance thermistors, including the Fluke 5611 probe; evaluated the resulting range and gain trade-offs.
- Demonstrated agreement within ± 1 mK between dual-ADC board temperatures and temperatures calculated from sensor resistance measured with an 8.5-digit Fluke multimeter under the tested temperature-chamber conditions; compared performance, cost, and size to help select a suitable board for each measurement point.
- Independently developed a bridge-board noise budget and LTspice model, predicting ~122 µV RMS output noise. Investigated an initial 2.68 mV noise measurement, identified 50 Hz mains and environmental interference as major contributors, and measured ~140 µV after power-isolation, grounding, and shielding improvements.
- Designed and revised board-level sensing and control circuits, including pressure acquisition, constant-current sources, digital I/O, and RS-422 interfaces; verified board operation and diagnosed failures through laboratory testing and on-site cleanroom troubleshooting.

Selected Circuit Project

Two-Stage CMOS Op-Amp Topology Comparison Jan. – May 2023

University of Florida course project | Team of two

- Designed and simulated three two-stage CMOS op-amps with a teammate in Cadence (conventional, folded-cascode, and telescopic-cascode; 1.8 V supply, 2 pF load). The telescopic design achieved 69 dB gain, 7 MHz unity-gain frequency, 79° phase margin, and 64.8 µW power—the lowest power and highest phase margin among the three designs; the folded design achieved the highest gain (87 dB).

Current Doctoral Research

PET Diffractive Waveguides for Upward-Gaze Medical AR Sep. 2026 – Present

Institute of Science Tokyo

- Completed preliminary grating-parameter design and Zemax OpticStudio simulations for PET-based diffractive waveguides using three one-dimensional gratings for in-coupling, folding, and out-coupling to achieve 2D exit-pupil expansion; currently refining the simulation model before the first fabrication run.
- Planned fabrication route: lithographic masters followed by roll-to-roll UV nanoimprint replication with an industry partner. Subsequent evaluation is intended to assess field of view, eyebox, optical efficiency, and brightness uniformity, with hospital-based validation planned for a later stage.

AR-Assisted Surgical Task and Visual Fatigue Study Sep. 2026 – Present

- Set up simulated heart-rate, blood-pressure, and SpO2 overlays on an existing AR headset for a sequential bead-placement task modeling surgical fine-motor work, examining access to upper-field information while preserving the central working view.
- Used eye tracking to check edge-of-display visibility and measured critical flicker fusion (CFF) before and after the task to investigate fatigue-related changes.

Selected Photonics Project

Grating Coupler Design for Silicon Photonics Dec. 2021 – Jun. 2022

FDTD simulation project

- Simulated and optimized an SOI grating coupler in Lumerical FDTD by varying period, duty cycle, and etch depth; obtained modeled coupling efficiencies of 43% (TE) and 13.3% (TM), with polarization dependence remaining a limitation.

Selected Publications

[1] Zhou, T., Zhang, G., and **Cai, Y.** "Unsupervised Autoencoders Combined with Multi-Model Machine Learning Fusion for Improving the Applicability of Aircraft Sensor and Engine Performance Prediction." *Optimizations in Applied Machine Learning*, 2025. DOI: 10.71070/oaml.v5i1.83

[2] Zhou, T., Zhang, G., and **Cai, Y.** "Residual Self-Attention-Based Temporal Deep Model for Predicting Aircraft Engine Failure within a Specific Cycle." *Optimizations in Applied Machine Learning*, 2023.

Technical Skills

Circuit design: Cadence (transistor-level), LTspice, MATLAB, Altium Designer

Programming: Python, C, Verilog HDL | Optics: Zemax OpticStudio, Lumerical FDTD

Laboratory: oscilloscopes, 8.5-digit multimeters, low-noise measurement, eye tracking

Board verification: power sequencing, ADC/DAC, serial interfaces, signal integrity, protection